or infundibuliform (*Petunia*) usually alternate with sepals; rarely bilipped and zygomorphic (*Schizanthus*) usually valvate, sometimes convolute (*Datura*).

Androecium: Stamens 5, epipetalous, filaments usually unequal in length, stamens only 2 in *Schizanthus*(others 3 are reduced to staminode), 4 and didynamous in (*Salpiglossis*) Anthers dithecous, dehisce longitudinally or poricidal.

Gynoecium: Bicarpellary, syncarpous obliquely placed, ovary superior, bilocular but looks tetralocular due to the formation of false septa, numerous ovules in each locule on axile placentation.

Fruit: A capsule or berry. In *Lycopersicon esculentum, Capsicum*, the fruit is a berry and in species of *Datura* and *Petunia*, the fruit is a capsule.

Seed: Endospermous.

Botanical description of *Datura metel*

Habit: Large, erect and stout herb.

Root: Branched tap root system.

Stem: Stem is hollow, green and herbaceous with strong odour.

Leaf: Simple, alternate, petiolate, entire or deeply lobed, glabrous exstipulate showing unicostate reticulate venation.

Inflorescence: Solitary and axillary cyme.

Flower: Flowers are large, greenish white, bracteate, ebracteolate, pedicellate, complete, heterochlamydeous, pentamerous, regular, actinomorphic, bisexual and hypogynous.

Calyx: Sepals 5, green synsepalous showing valvate aestivation. Calyx is mostly persistant, odd sepal is posterior in position.

Corolla: petals 5, greenish white, sympetalous, plicate (folded like a fan) showing twisted aestivation, funnel shaped with wide mouth and 10 lobed.

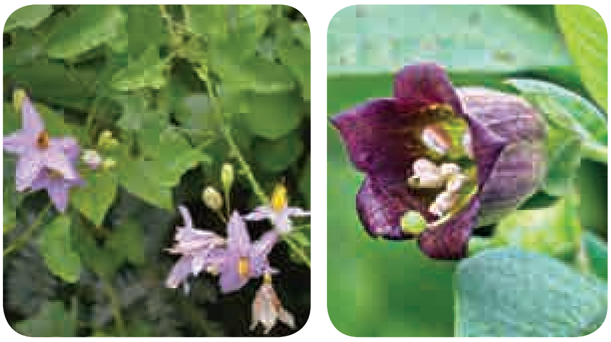
Androecium: Stamens 5, free from one another, epipetalous, alternipetalous and are inserted in the middle of the corolla tube. Anthers are basifixed, dithecous, with long filament, introse and longitudinally dehiscent.

Gynoecium: Ovary bicarpellary, syncarpous superior ovary, basically biloculear but tetralocular due to the formation of false septum. Carpels are obliquely placed and ovules on swollen axile placentation. Style simple long and filiform, stigma two lobed.

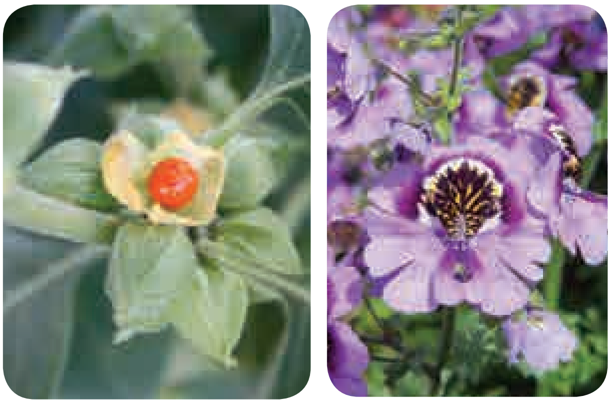
Fruit: Spinescent capsule opening by four apical valves with persistent calyx.

Seed: Endospermous.

Floral Formula: Dt0Gdtn0⊕.⚥.$M_{*7+}.\overset{\frown}{E_{*7+}.C_{7}}.\underline{I}_{*4+}$



Solanum trilobatum *Atropa belladonna*



Withania somnifera *Schizanthus pinnatus*

Figure 5.19: Selected plants belongs to the Family Solanace